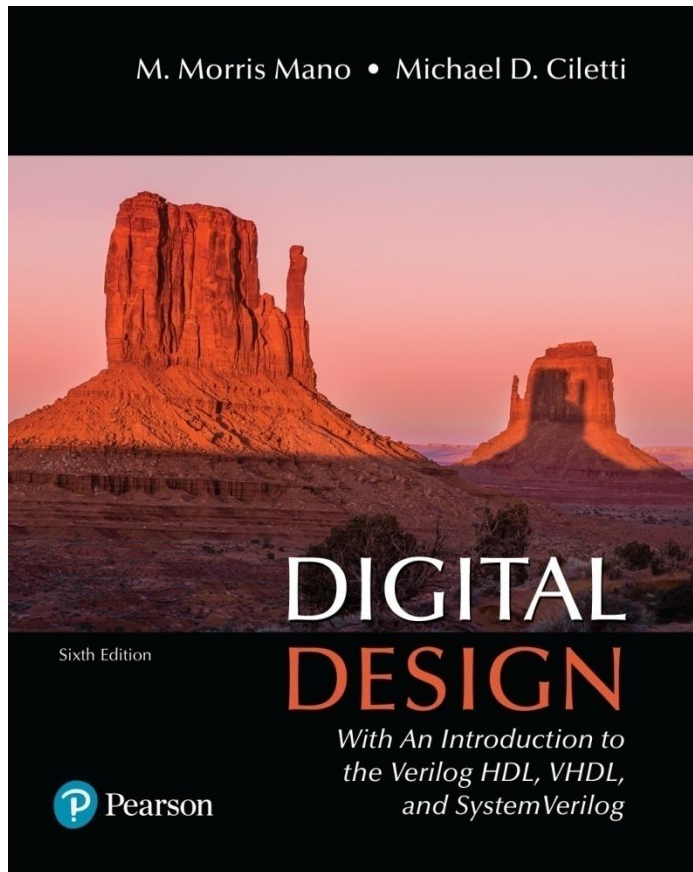


Digital Design

With an Introduction to the Verilog HDL, VHDL, and SystemVerilog

6th Edition



Chapter 10

Standard Graphic Symbols

FIGURE 10.1

Rectangular-shape graphic symbols for gates.

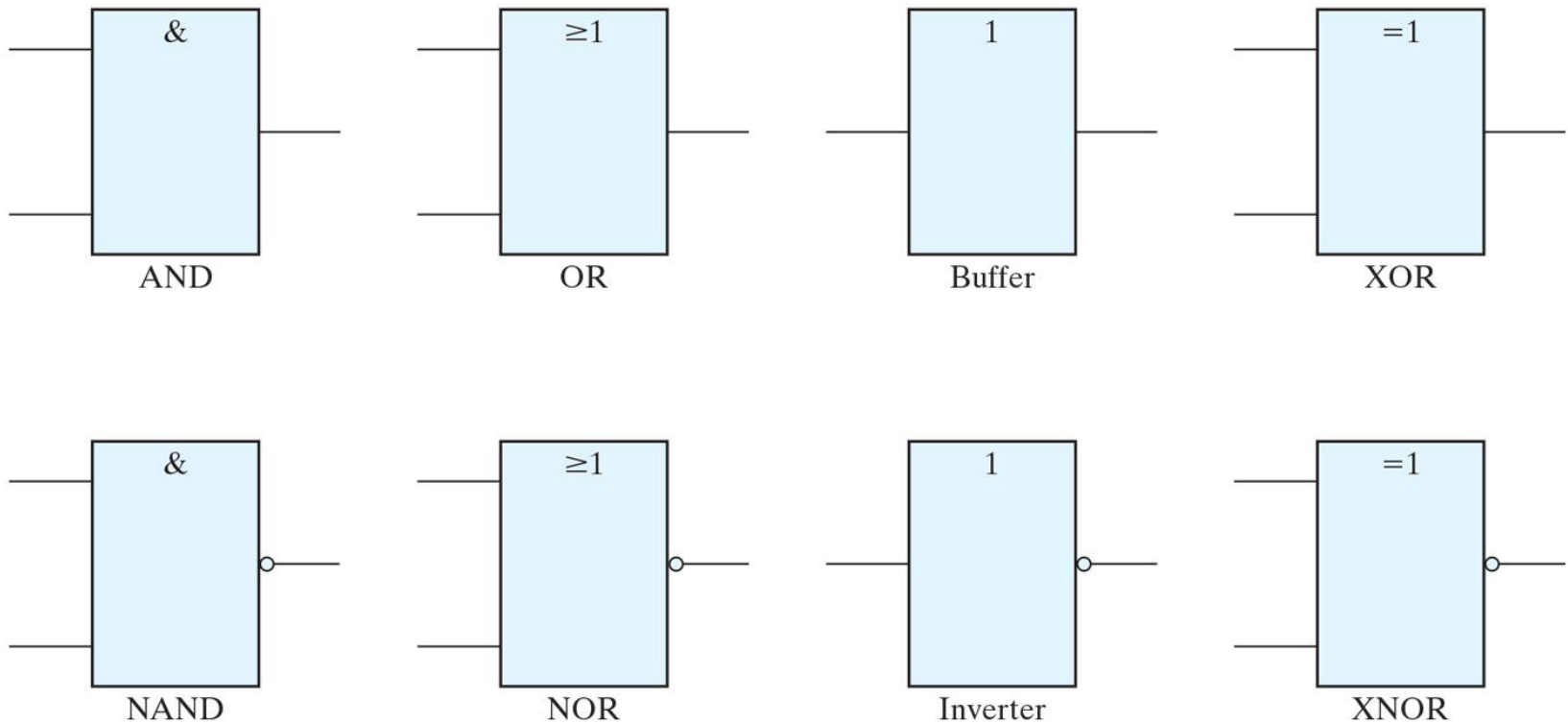


FIGURE 10.2

Standard graphic symbol for a four-bit parallel adder, IC type 7483.

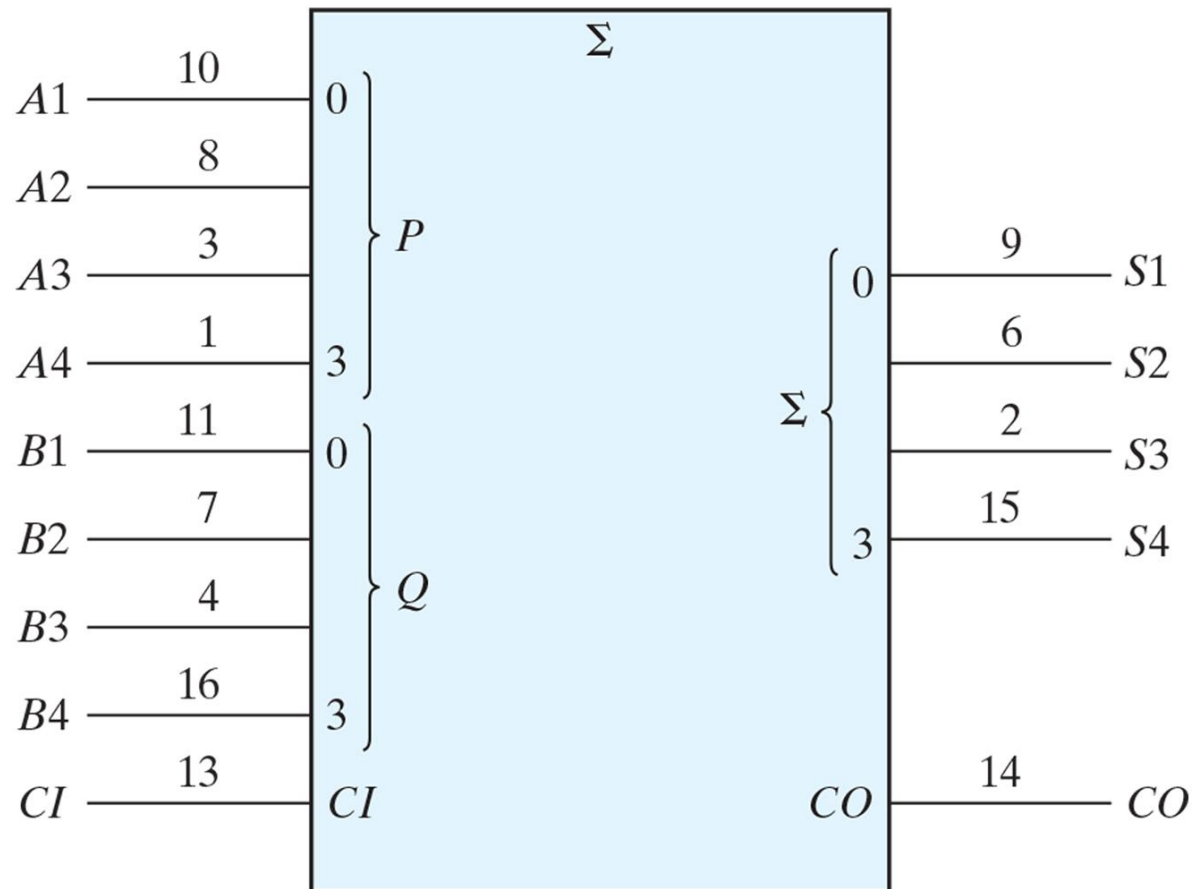


FIGURE 10.3

Standard graphic symbol for a 2-to-4-line decoder (one-half of IC type 74155).

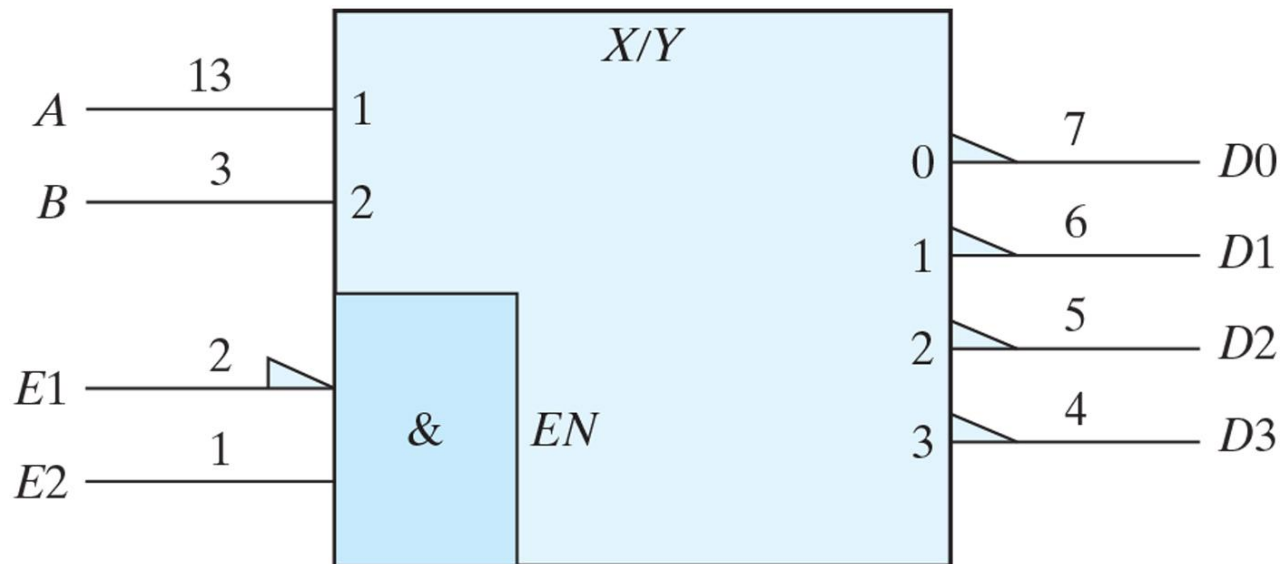


Table 10.1
General Qualifying Symbols.

Symbol	Description
&	AND gate or function
≥ 1	OR gate or function
1	Buffer gate or inverter
$= 1$	Exclusive-OR gate or function
2k	Even function or even parity element
$2k + 1$	Odd function or odd parity element
X/Y	Coder, decoder, or code converter
MUX	Multiplexer
DMUX	Demultiplexer
Σ	Adder
Π	Multiplier
COMP	Magnitude comparator
ALU	Arithmetic logic unit
SRG	Shift register
CTR	Counter
RCTR	Ripple counter
ROM	Read-only memory
RAM	Random-access memory

FIGURE 10.4

Qualifying symbols associated with inputs and outputs.

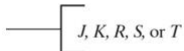
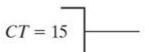
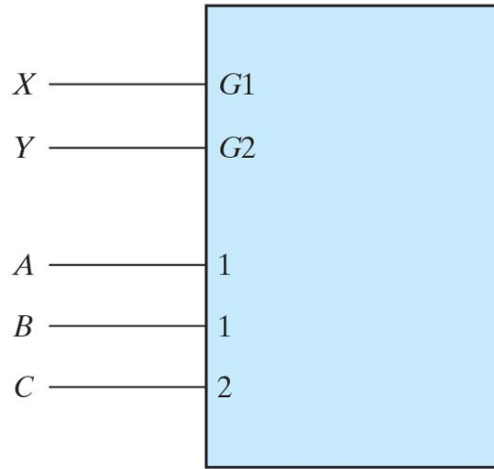
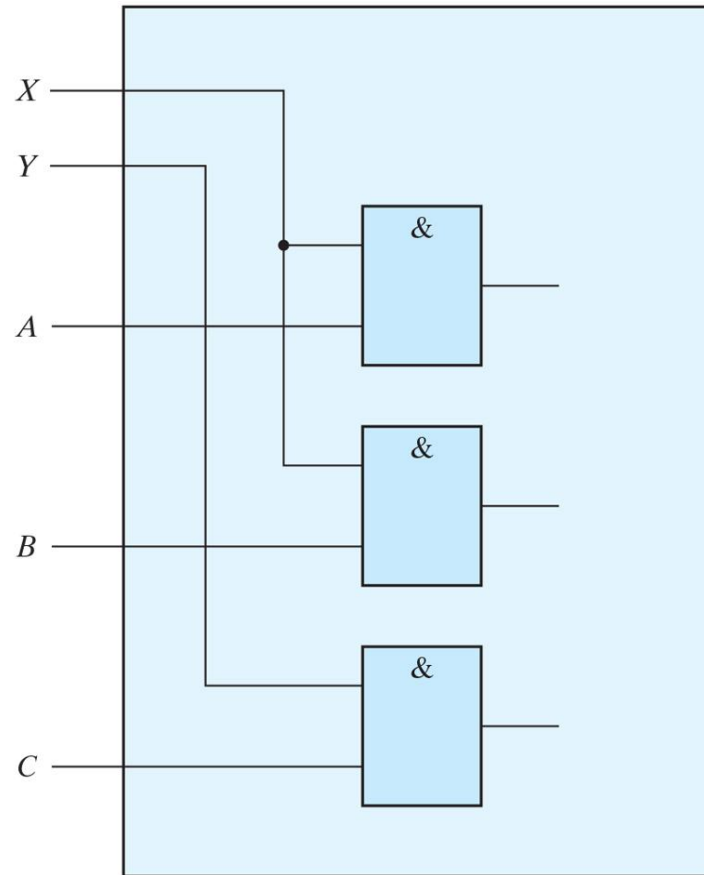
Symbol	Description
	Active-low input or output
	Logic negation input or output
	Dynamic indicator input
	Three-state output
	Open-collector output
	Output with special amplification
	Enable input
	Data input to a storage element
	Flip-flop inputs
	Shift right
	Shift left
	Countup
	Countdown
	Contents of register equals binary 15

FIGURE 10.5

Example of G (AND) dependency.



(a) Block with $G1$ and $G2$



(b) Equivalent interpretation

FIGURE 10.6

IC type 74155 connected as a 3 × 8 decoder.

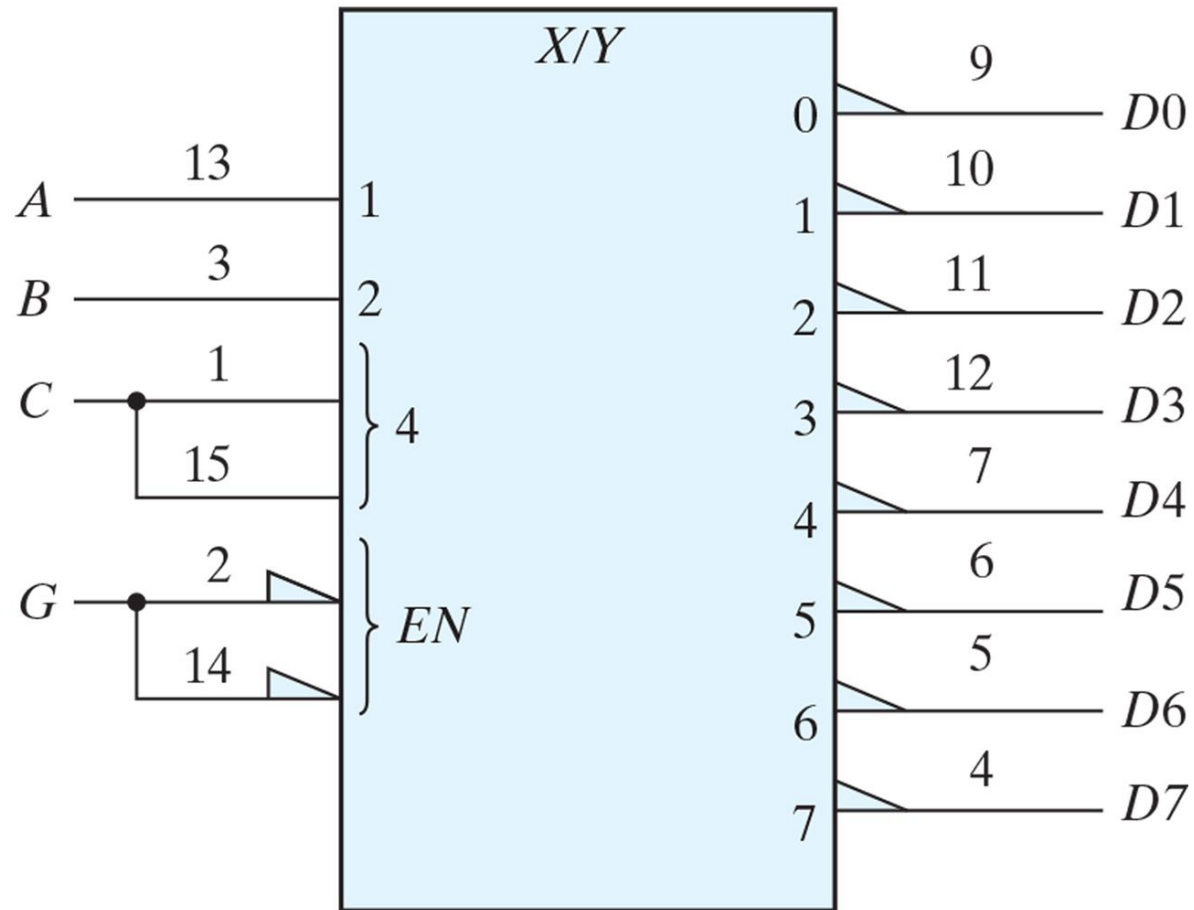


FIGURE 10.7

Graphic symbols for multiplexers.

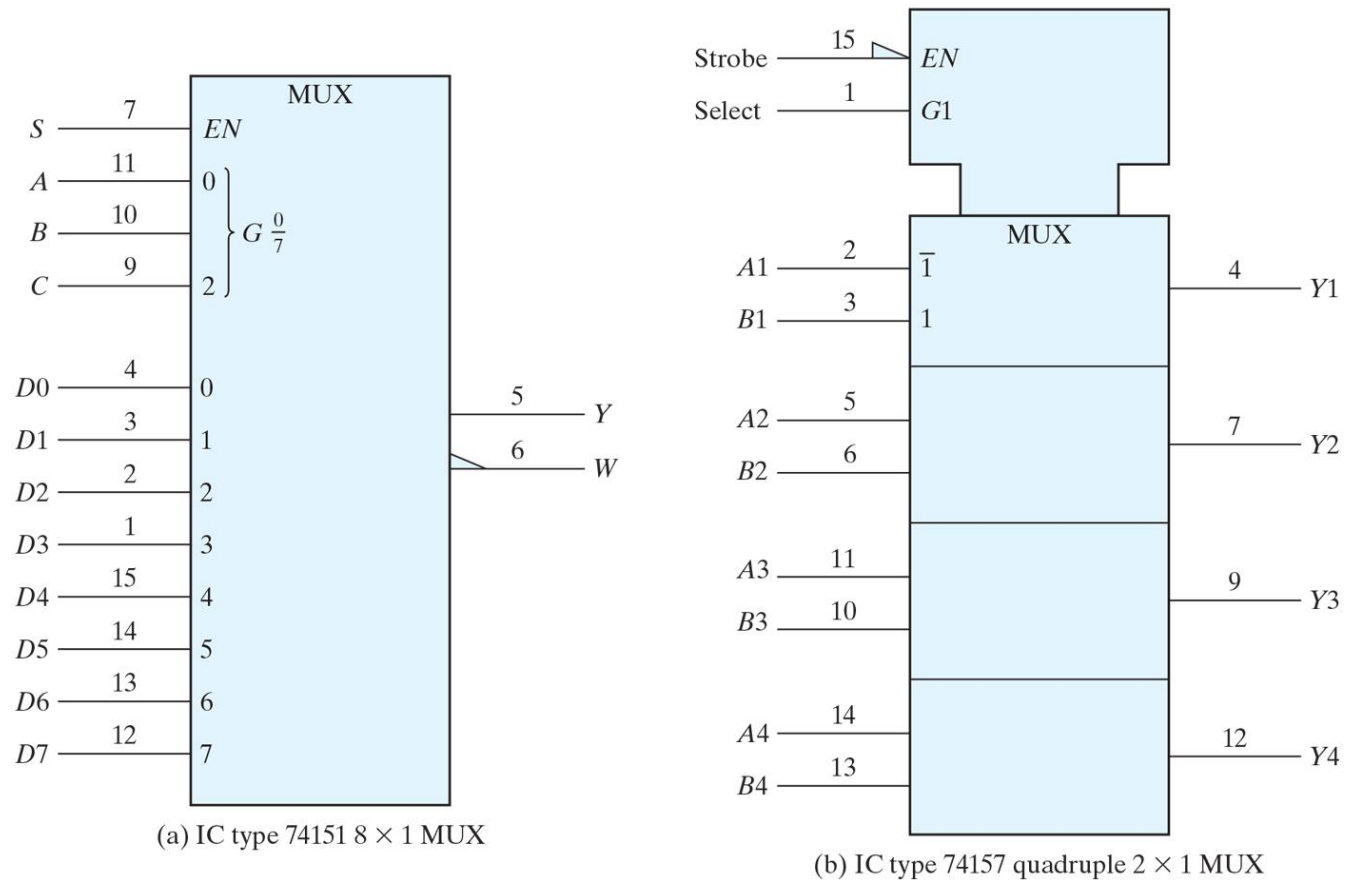


FIGURE 10.8

Standard graphic symbols for flip-flops.

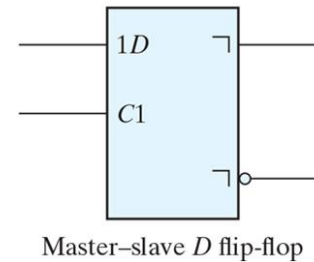
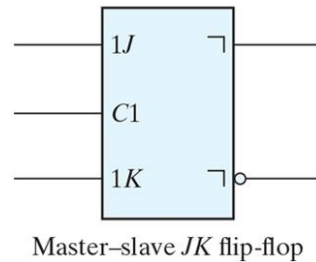
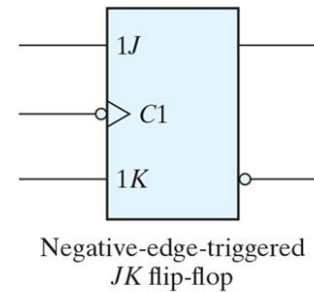
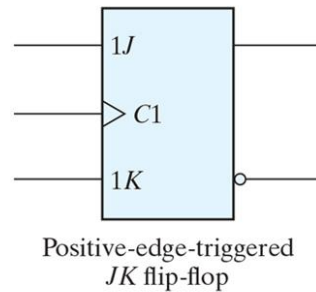
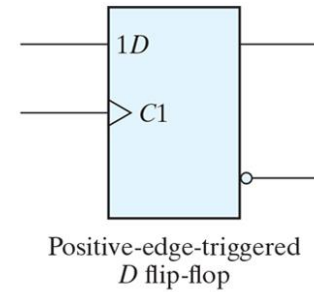
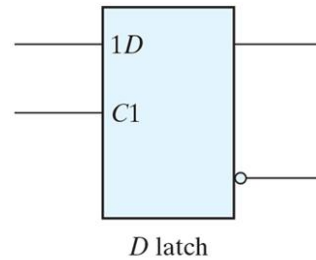
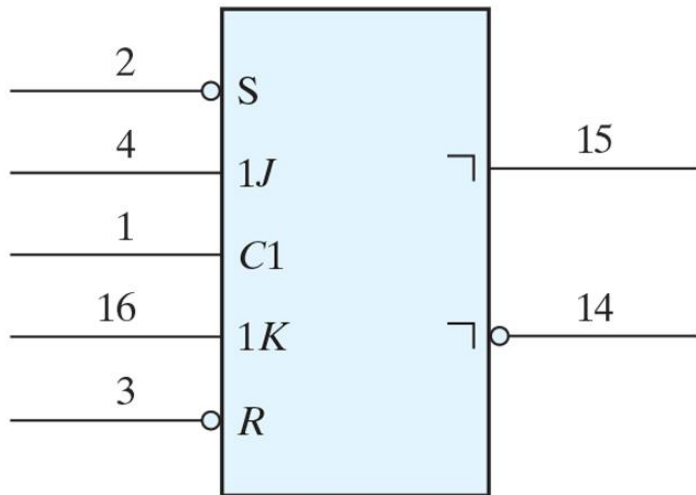
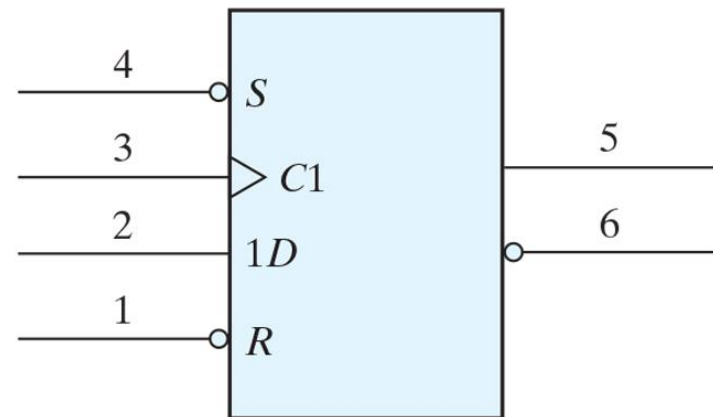


FIGURE 10.9

IC flip-flops with direct set and reset.



(a) One-half 7476 *JK* flip-flop



(b) One-half 7474 *D* flip-flop

FIGURE 10.10

Graphic symbol for IC type 74175 quad flip-flop.

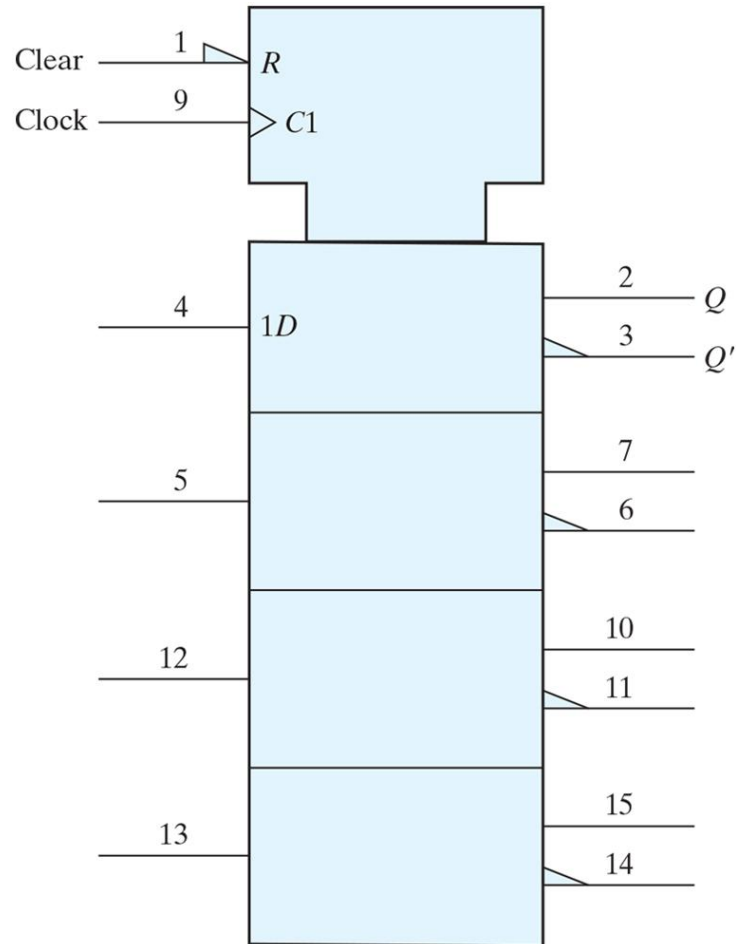


FIGURE 10.11

Graphic symbol for a shift register with parallel load, IC type 74195.

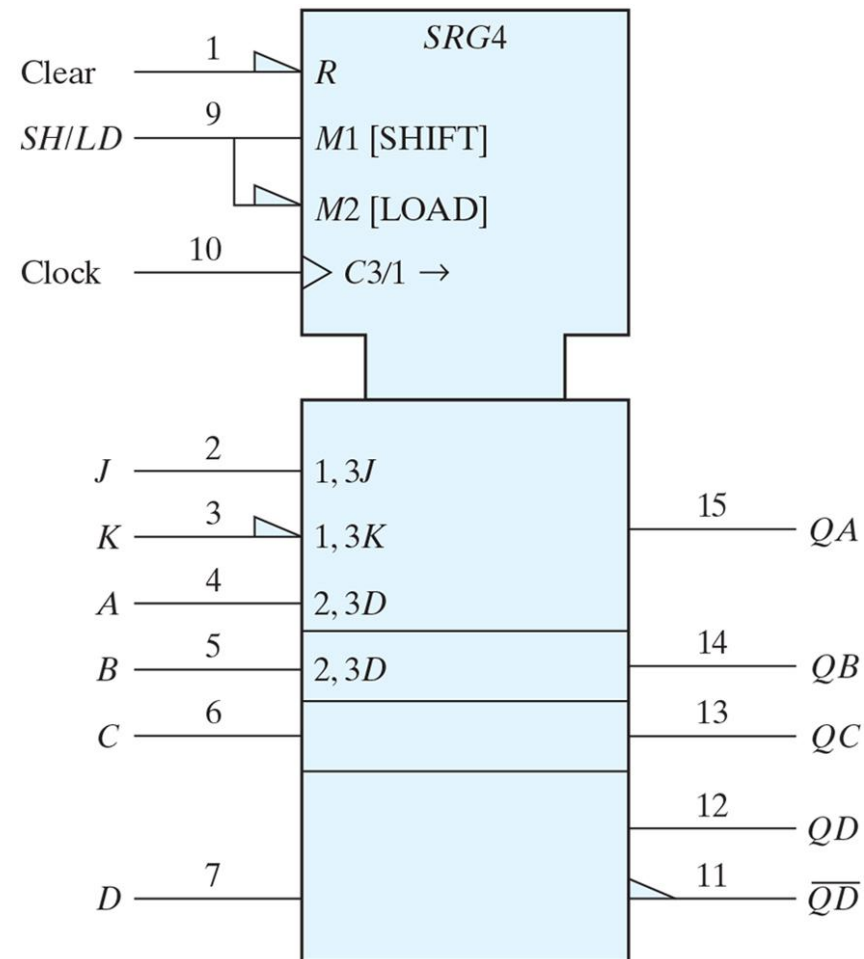


FIGURE 10.12

Graphic symbol for a bidirectional shift register with parallel load, IC type 74194.

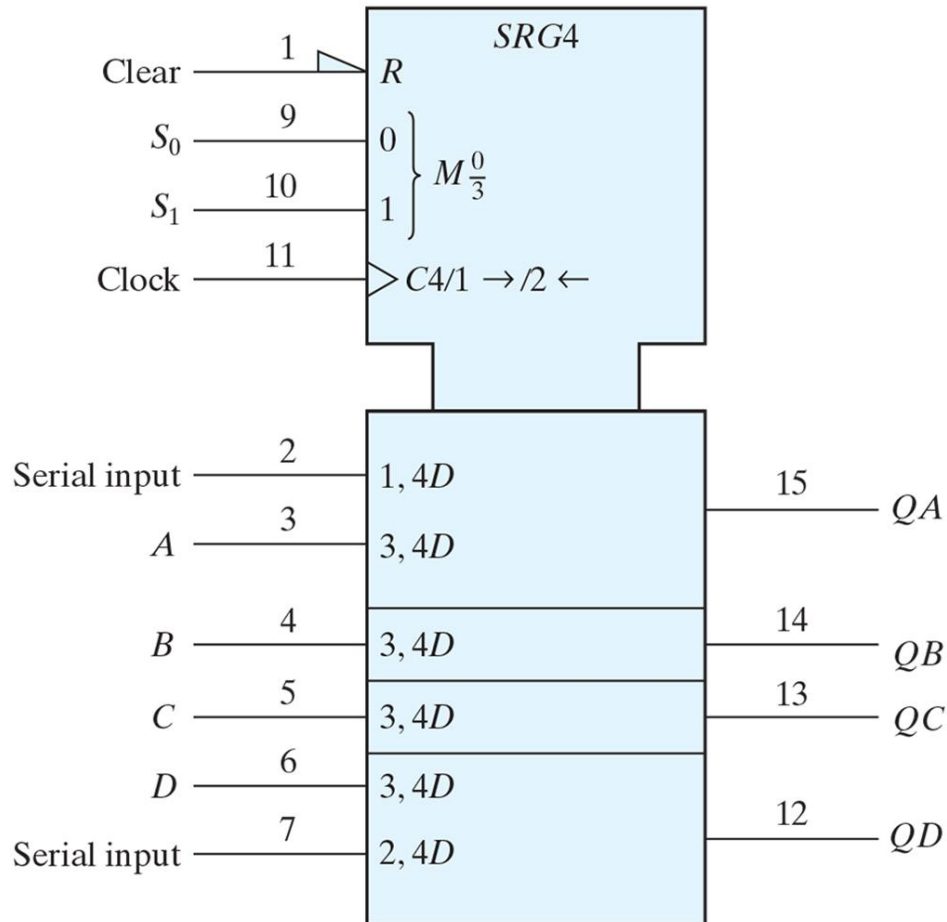


FIGURE 10.13

Graphic symbol for ripple counter, IC type 7493.

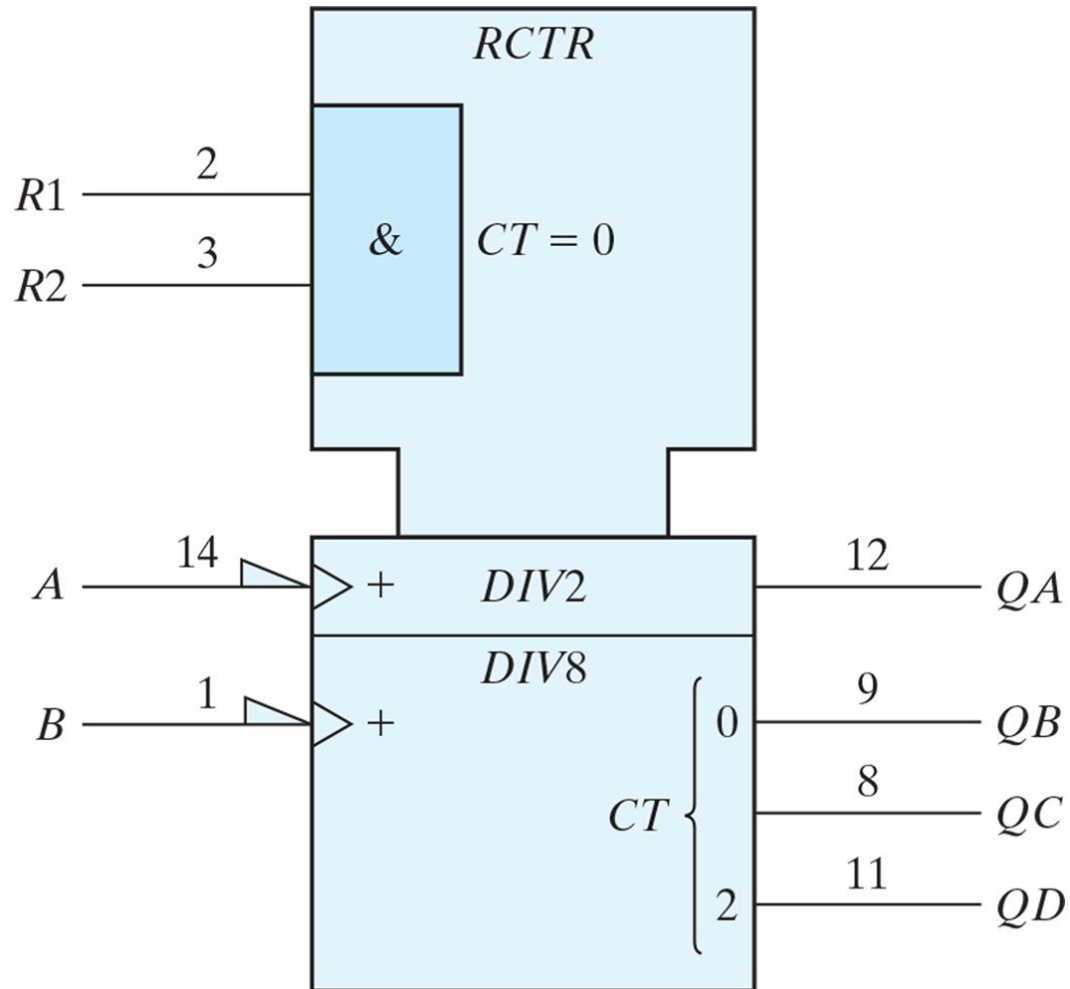


FIGURE 10.14

Graphic Symbol for 4-Bit Binary Counter with Parallel Load, IC Type 74161.

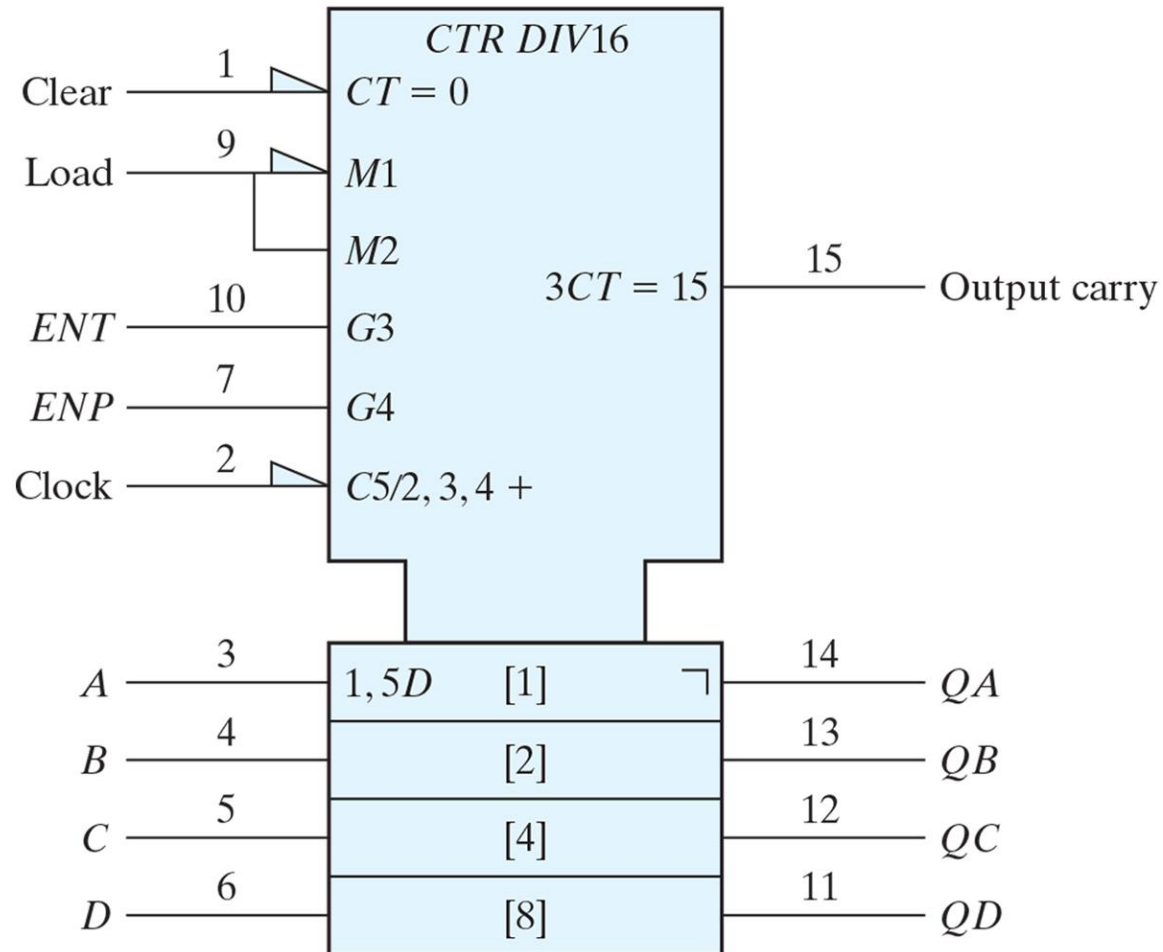


FIGURE 10.15

Graphic symbol for 16×4 RAM, IC type 74189.

